

Applied Machine Learning

Unit 01 — Lecture 01 Notes

Introduction to Machine Learning & the Python ML Stack

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Read this first

These notes are written so that you can learn the material *without* having been in the lecture. Every concept is developed from scratch, every number you see was produced by running the code shown beside it, and every figure was generated from real computation rather than drawn by hand.

Work through them with a Python session open. Sections marked **Try it yourself** are the point of the whole document — reading about machine learning teaches you roughly as much as reading about swimming.

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1 How to use these notes

Read in order. The sections build on each other, and Section 4 in particular will not make sense before Section 3.

There are three kinds of box:

- **Key idea** — the one sentence from that section you should be able to reproduce a month later.
- **Worked example** — a calculation done by hand, with small numbers, before any library is involved. Do these with a pen. If you can do them by hand, the library will never be a black box.
- **Try it yourself** — something to run. The expected output is given so you know whether it worked.

Code blocks with a **grey** bar to the left are *output*, not input:

```
print(2 + 2)
```

4

Every exercise in Section 9 has a full solution in Section 10. Attempt them before reading the solutions; the answers will not stick otherwise.

What you need installed

```
python -m venv aml-env
source aml-env/bin/activate # Windows: aml-env\Scripts\activate
pip install numpy pandas scikit-learn matplotlib seaborn jupyter
```

The numbers in these notes were produced with Python 3.10, NumPy 2.2, pandas 2.3, scikit-learn 1.7 and matplotlib 3.10. Small differences in your versions will change the last decimal place, not the conclusions.

Learning outcomes

By the end of these notes you should be able to:

1. Explain what machine learning is, and recognise problems where it is *not* the right tool.
2. State Mitchell's definition and apply it: given a system, name its task T , performance measure P and experience E .
3. Distinguish supervised, unsupervised and reinforcement learning, and classify an unfamiliar problem into one of them.
4. Explain why fitting the training data is easy and why generalisation is the real goal, and describe how a train/test split and cross-validation estimate it.
5. Recognise data leakage and explain why it inflates a score.
6. Use NumPy, pandas and scikit-learn to load data, inspect it, and fit and evaluate a model in a dozen lines.

Mapped to **CO1** (core concepts and techniques of ML and AI) and **CO2** (develop models using popular libraries and frameworks).

2 What machine learning is

2.1 The shape of the problem

Some programs we can write by hand because we know the rule. Sorting a list. Parsing a date. Computing a payroll. We state the rule, we encode the rule, and the program is correct by construction.

Other programs we cannot write by hand at all. Decide whether an email is spam. Read a handwritten digit. Tell whether a chest scan shows a tumour. You can *recognise* the right answer in each case — you would know spam if you saw it — but you cannot write down the rule that produces the answer. Try it: write a precise, complete rule that distinguishes the handwritten digit 3 from the digit 8, covering every handwriting style in this room. You will not manage it, and neither has anyone else.

Key idea

Machine learning is what you reach for when you have **examples of the answer** but not **a statement of the rule**. Instead of writing the rule, you write a program that *finds* the rule from the examples.

That framing describes the *supervised* case, which is most of this course and most industrial machine learning. Section 3 introduces two other families that relax it: one where you have no answers at all, and one where you get scored on your actions rather than told the answer.

2.2 A first contrast: two ways to build a spam filter

Suppose you must filter spam. The pre-machine-learning approach is to write conditions:

```
def classify(subject, sender, exclamation_marks):
    if "free money" in subject: return "SPAM"
    if sender in blocklist: return "SPAM"
    if exclamation_marks > 5: return "SPAM"
    return "HAM"
```

This works on Monday. On Tuesday the spammer writes `fr33 m0ney` and you add a rule. On Wednesday they pad the message with ordinary words and you add another. You are now in an arms race in which you personally are the bottleneck, because every adaptation by the adversary requires a code change by you.

The learned filter looks like this:

```
model.fit(emails, labels) # learn from examples
model.predict(new_email) # apply to something new
```

When the spammer adapts, you do not read their new trick and encode it. You collect newly labelled mail — users pressing the “spam” button generate it for free — and refit. What has changed is not only the code but the **maintenance strategy**: adaptation became a data problem instead of a programming problem.

What you gave up

The rule-based filter has one property the learned one loses: you can read exactly why it fired. “Blocked because the subject contained *free money*” is an explanation a user can act on. A learned model’s answer is a number produced by weights, and recovering an explanation from it is a research area in itself. That property is called **interpretability**,

and Unit VI returns to it when we meet decision trees, which are learned *and* readable.

2.3 A working definition

The definition every ML course starts from is Tom Mitchell's, from his 1997 textbook:

A computer program is said to **learn** from experience E with respect to some class of tasks T and performance measure P , if its performance at tasks in T , as measured by P , improves with experience E .

It is worth slowing down here, because the definition is doing more work than it appears to. It insists you name three things before you have a machine learning problem at all. Most of the difficulty in real projects is that somebody skipped one of them.

Naming T

For each system below, name the task, the experience and the performance measure. Cover the right-hand columns and try it before reading on.

System	Task T	Experience E	Performance P
Spam filter	classify an email as spam or not	a labelled inbox	fraction of emails classified correctly
House-price estimator	predict the sale price	records of past sales with their prices	mean absolute error, in rupees
Chess engine	choose a move	games played against itself	fraction of games won

Notice that P is the hardest column. "Fraction correct" sounds obvious for the spam filter until you realise that a filter which passes everything through is 97% correct if only 3% of your mail is spam. We return to this in Unit VI.

Key idea

The definition forces you to name P **before** you start. A great many machine-learning projects fail here rather than at the modelling step: nobody agreed in advance what "working" would mean, so there was no way to tell whether the finished system worked.

2.4 Where machine learning sits

Three terms get used interchangeably in the press and they are not the same thing.

- **Artificial intelligence** is the broad field: search, planning, logic, knowledge representation, robotics, and learning. A chess engine using pure search with no learning at all is AI.
- **Machine learning** is the part of AI that learns from data or experience rather than being programmed with the rule.
- **Deep learning** is a part of machine learning that uses neural networks with many layers.

So $\text{AI} \supset \text{ML} \supset \text{DL}$.

Deep learning is not an upgrade

Deep learning is a *subset*, not a synonym and not a strictly better option. On tabular data of the size you will meet in this course — thousands to hundreds of thousands of rows, tens of columns — a gradient-boosted tree usually matches or beats a neural network, trains in seconds rather than hours, and needs far less tuning. Choosing the smallest model that solves the problem is a professional skill, not a compromise.

2.5 Why now?

Most of the mathematics in this course is old. Least squares, which you meet in Unit V, was published by Legendre in 1805. The perceptron dates from 1958. Backpropagation was in use in the 1980s. So why is machine learning suddenly everywhere?

- **Data.** Transactions, sensors, images, clicks and logs are all digital and cheap to store. The examples that a learning algorithm needs now exist as a by-product of ordinary business.
- **Compute.** Parallel hardware made the arithmetic in Unit III affordable at a scale that was impossible in 1990.
- **Algorithms and tooling.** Better optimisers, better regularisation, and libraries that put all of it three lines away.

The third point matters more than students expect. Much of what you will do in this course would have been a PhD project in 1995 and is now an import statement. That is a real change in what one person can accomplish, and it is why the course spends its time on *judgement* — which method, which metric, is this result honest — rather than on implementation.

2.6 When *not* to use machine learning

Knowing when a tool does not apply is as much a part of competence as knowing how to use it. Do not reach for machine learning when:

1. **The rule is known and stable.** Income tax is a published formula. Implement the formula. A model that predicts tax with 99% accuracy is strictly worse than arithmetic that gets it right every time.
2. **You have no labelled examples** and no realistic way to obtain them. No examples, no supervised learning — and collecting labels is usually the most expensive part of a project.
3. **Errors are unacceptable and you cannot bound them.** A learned model gives you a rate, not a guarantee. If a single failure is catastrophic and unbounded, you need a different kind of system.
4. **Every decision must be explained** to a regulator or a court, and a simple rule would do nearly as well. Loan refusals and medical decisions live here.
5. **The training data will not resemble deployment.** A model trained on last year's customers predicts last year's customers. If the world has moved, so has the answer.

Think before the next session

A hospital wants to predict which patients will miss an appointment, so it can overbook. Which of the five cautions above is most likely to bite, and why? Write three or four sentences. The answer is in Section 10, Exercise **Q2**..

3 The three families of machine learning

3.1 The question that decides which family you are in

Everything in this section follows from one question:

Key idea

Do you already have the answers?

Yes, for past examples → **supervised** learning.

No, and you want to find structure → **unsupervised** learning.

No, but you can act and be scored → **reinforcement** learning.

Family	What you have	What you learn
Supervised	labelled pairs $\{(x_i, y_i)\}$	a mapping $x \mapsto y$
Unsupervised	unlabelled data $\{x_i\}$	structure within x
Reinforcement	an environment and a reward	a policy: what to do when

3.2 Supervised learning

You are given n examples $\{(x_1, y_1), \dots, (x_n, y_n)\}$, where each x_i is a vector of **features** (the inputs) and each y_i is the **label** or **target** (the known answer). The goal is to find a function f such that $f(x) \approx y$ — crucially, on data you have *not* seen.

Supervised problems split in two according to what y is:

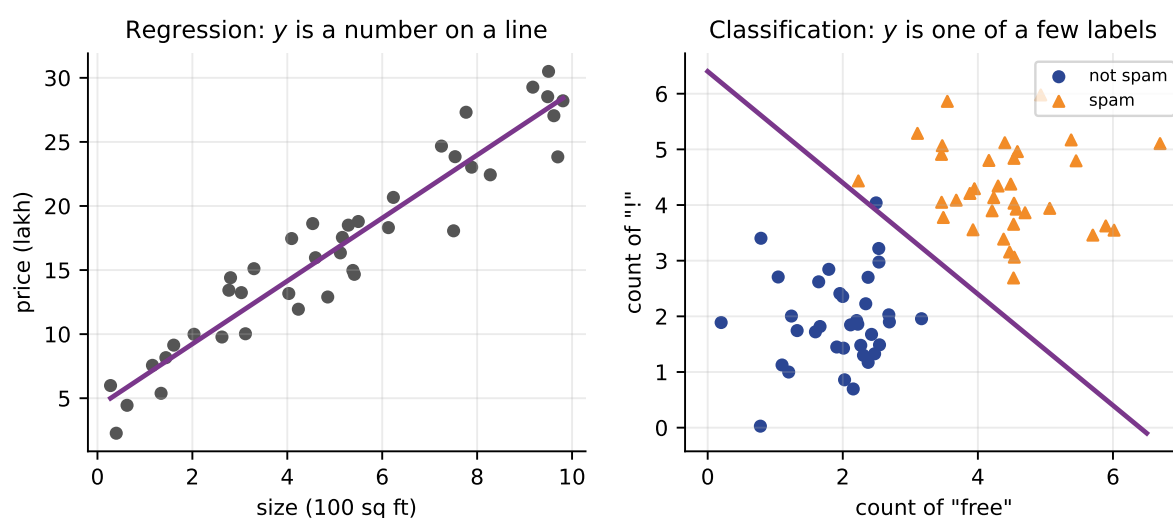


Figure 1: Left: regression, where the target is a number and the model draws a line (or surface) through the data. Right: classification, where the target is one of a few labels and the model draws a boundary between them.

- **Regression** — y is a continuous number. House price, stock value, tomorrow's temperature, hours until a machine fails. *Unit V; lab experiments 3–4.*
- **Classification** — y is one of a finite set of categories. Spam or not spam; digit 0–9; benign or malignant. *Unit VI; lab experiments 5–9 and 11–12.*

The distinction is about the *target*, not about the features or the method. The same features

can support both: given a patient’s measurements you might predict their blood pressure (regression) or whether they are hypertensive (classification).

3.3 A supervised algorithm you can run in your head

Before any library, here is a complete supervised algorithm. ***k*-nearest neighbours** makes a prediction for a new point by finding the *k* training points closest to it and taking a vote.

1-nearest neighbour

Five training flowers, described by (petal length, petal width) in cm:

petal length	petal width	species
1.4	0.2	setosa
1.3	0.2	setosa
4.7	1.4	versicolor
4.5	1.5	versicolor
6.0	2.5	virginica

A new flower measures (4.6, 1.4). Which species does 1-NN predict?

Compute the Euclidean distance $\sqrt{(a_1 - b_1)^2 + (a_2 - b_2)^2}$ to each:

Training point	Working	Distance
(1.4, 0.2)	$\sqrt{3.2^2 + 1.2^2} = \sqrt{10.24 + 1.44}$	3.42
(1.3, 0.2)	$\sqrt{3.3^2 + 1.2^2} = \sqrt{10.89 + 1.44}$	3.51
(4.7, 1.4)	$\sqrt{0.1^2 + 0.0^2} = \sqrt{0.01}$	0.10
(4.5, 1.5)	$\sqrt{0.1^2 + 0.1^2} = \sqrt{0.02}$	0.14
(6.0, 2.5)	$\sqrt{1.4^2 + 1.1^2} = \sqrt{1.96 + 1.21}$	1.78

The nearest is (4.7, 1.4) at distance 0.10, so 1-NN predicts **versicolor**. With *k* = 3 the three nearest are versicolor, versicolor and virginica — the vote is 2–1, and the answer is versicolor again.

Two things to notice. First, this algorithm has no “training” in any meaningful sense: it just stores the data and does all its work at prediction time. Second, the whole method rests on *distance*, which is why Section 7 makes such a fuss about the scale of your features.

Here is the same computation in NumPy, so you can check your arithmetic:

```
import numpy as np
train = np.array([[1.4,0.2],[1.3,0.2],[4.7,1.4],[4.5,1.5],[6.0,2.5]])
new = np.array([4.6, 1.4])
d = np.sqrt(((train - new)**2).sum(axis=1))
print(np.round(d, 2))
print("nearest row:", d.argmin())
```

```
[3.42 3.51 0.1 0.14 1.78]
nearest row: 2
```

3.4 Unsupervised learning

Here there is no *y* at all. You have observations and want to know what structure is in them.

- **Clustering** — group similar records. Customer segments, disease subtypes, document topics. *Unit VII; lab experiment 13.*
- **Dimensionality reduction** — describe the data with fewer features while keeping most of the information. *Unit IV.*

- **Anomaly detection** — identify observations that do not fit the pattern. Fraud, equipment faults. *Lab experiment 10.*

Unsupervised results are harder to judge

With no ground truth there is nothing to compute an accuracy against. If a clustering splits your customers into four groups, no one can tell you whether those are the “right” four groups — there is no right. Evaluating a clustering is genuinely harder than evaluating a classifier, and Unit VII spends real time on it.

3.5 Reinforcement learning

An agent takes actions in an environment and receives a reward. It is never told the correct action; it only observes consequences, and those consequences are often delayed — a chess move’s value may not be apparent for thirty moves. Used in robotics, game playing, control systems and recommendation with feedback. We cover it here only so you can recognise it; it is not examined beyond the distinction.

3.6 Naming the problem

The skill this section is really teaching is classifying an unfamiliar problem. Cover the two right-hand columns and try each row.

Situation	Family	Task
10 000 past loans marked repaid or defaulted; score a new applicant	supervised	classification
Five years of daily sales; forecast next month	supervised	regression
50 000 customer records, no labels; find natural groups	unsupervised	clustering
Card transactions; flag the unusual ones	unsupervised	anomaly detection
A warehouse robot that must learn a route	reinforcement	policy learning

Row four repays thought. Fraud detection *could* be supervised — if you had a reliable label for every past transaction. In practice you do not: confirmed fraud is rare, arrives months late, and is itself biased by whatever detection system was running at the time. So a problem that is supervised in principle is often unsupervised in practice, and the reason is operational rather than mathematical.

4 The central difficulty: generalisation

4.1 Fitting the training data is trivial

Here is a model that achieves 100% accuracy on any training set: store every training example, and when asked to predict, look up the answer. Perfect score, zero learning. It has memorised, and it will fail completely on anything it has not seen.

Key idea

The goal is never to fit the data you have. It is to be right about data you have never seen. That property is called **generalisation**, and everything in the rest of this course — splits, cross-validation, regularization — is machinery for estimating it or improving it.

4.2 Holding data back

Since we cannot see the future, we simulate it: hide some of the data from the model, then use that hidden part to estimate how it will behave on genuinely new cases.

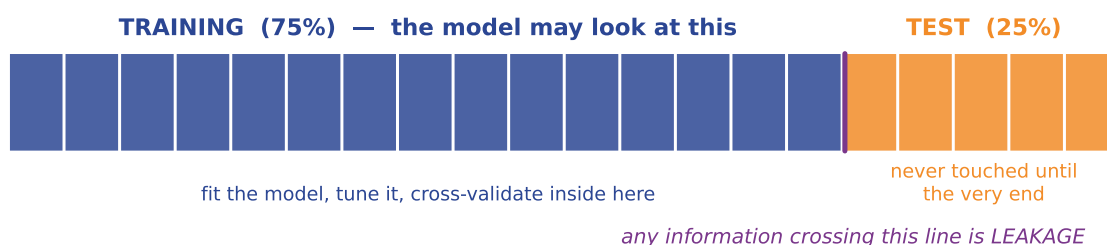


Figure 2: The train/test split. The model may look at the blue part as often as it likes. The orange part is opened once, at the very end, to produce the number you report.

The test set is not a tuning signal

If you try twenty models and pick the one with the best test score, that score is no longer an honest estimate — you have used the test set to make a choice, so it has leaked into your decision. The honest procedure is to split three ways (train / validation / test), tune on validation, and touch the test set once. Cross-validation, below, is a cheaper version of the same idea.

4.3 Overfitting, seen properly

Let us watch a model overfit. We generate 30 points from a gentle curve plus noise, hold out 40%, and fit polynomials of increasing degree.

The measured numbers behind that right-hand panel:

Model	Training RMSE	Test RMSE
Degree 1 (straight line)	1.32	1.60
Degree 2 (parabola)	0.89	1.14
Degree 3 (best test error)	0.86	1.10
Degree 12	0.57	40.52

Read the last row carefully. The degree-12 model is the *best* model by training error and catastrophically the worst by test error — off by 40 units where the data spans about 10. If

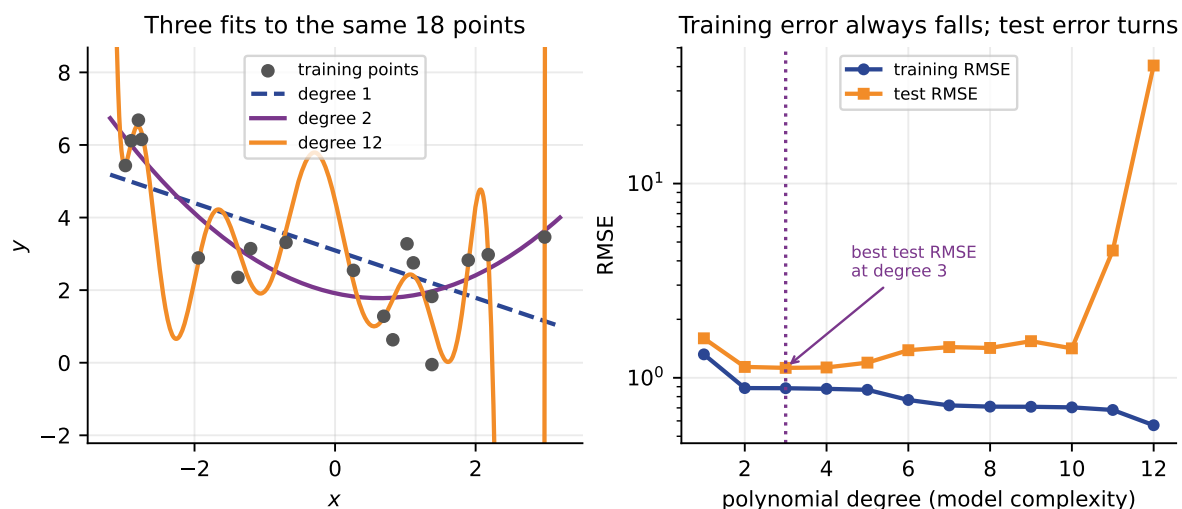


Figure 3: Left: a straight line under-fits, a parabola fits well, and a degree-12 polynomial contorts itself through every training point. Right: the same experiment measured. Training error falls monotonically as complexity increases; test error falls, bottoms out, then explodes.

you had judged it on training error you would have shipped the worst possible choice.

Key idea

The gap between training and test error is the diagnostic. A small gap with both errors high means under-fitting: the model is too simple. A tiny training error with a large test error means over-fitting: the model has learned the noise. You want the point where test error is lowest, which is usually *not* where training error is lowest.

4.4 Cross-validation

A single split has a problem: with a small dataset, which rows happen to land in the test set moves the answer around. Iris has 150 rows, so a 25% test set is 38 flowers; two flowers changing sides moves accuracy by five points.

k -fold cross-validation fixes this. Split the data into k equal folds; train on $k - 1$ of them and test on the one left out; repeat k times so each fold serves as the test set once; average.

```
from sklearn.model_selection import cross_val_score
scores = cross_val_score(model, X, y, cv=10)
print(scores.round(3))
print(f"mean {scores.mean():.3f} sd {scores.std():.3f}")
```

```
[1.  0.933 1.  0.933 0.867 0.933 0.867 1.  1.  1. ]
mean 0.953 sd 0.052
```

Look at the spread: individual folds range from 0.867 to 1.000. Reporting “97.4% accuracy” from one lucky split would have been true and misleading. The honest summary is 0.953 ± 0.052 .

Quote the spread

A single accuracy figure with no measure of variability is nearly meaningless on a small dataset. Get into the habit of reporting mean and standard deviation across folds, and of not quoting more significant figures than your spread justifies. “95%” is honest here; “95.33%” is not.

4.5 Data leakage

Data leakage is the use, at training time, of information that would not be available at prediction time. It is the single most common way to get a wonderful result that means nothing, and it is subtle enough that it catches professionals.

The clearest demonstration uses data with *no signal at all*. We generate 100 rows of 5000 pure-noise features and assign labels by coin flip. Nothing is predictable; the honest accuracy of any model is 50%.

```
import numpy as np
from sklearn.feature_selection import SelectKBest, f_classif
from sklearn.model_selection import cross_val_score
from sklearn.neighbors import KNeighborsClassifier
from sklearn.pipeline import make_pipeline

rng = np.random.default_rng(0)
X = rng.normal(size=(100, 5000)) # pure noise
y = rng.integers(0, 2, 100) # coin flips

# WRONG: pick the 20 "best" features using every row, including the test rows
Xsel = SelectKBest(f_classif, k=20).fit_transform(X, y)
wrong = cross_val_score(KNeighborsClassifier(3), Xsel, y, cv=10).mean()

# RIGHT: selection lives inside the pipeline, so it happens inside each fold
pipe = make_pipeline(SelectKBest(f_classif, k=20), KNeighborsClassifier(3))
right = cross_val_score(pipe, X, y, cv=10).mean()

print(f"selection before the split : {wrong:.2f}")
print(f"selection inside the fold : {right:.2f}")
```

```
selection before the split : 0.81
selection inside the fold : 0.50
```

Eighty-one percent accuracy on random noise. The feature selector examined all 5000 columns *including the test rows' labels* and picked the twenty that happened to correlate with y by chance. Those twenty then looked predictive during cross-validation, because the selection had already peeked.

Key idea

Any step that *learns something from the data* — scaling, imputation, feature selection, encoding — must be fitted on the training portion only. The practical rule that makes this automatic is: **put every such step inside a Pipeline** and let scikit-learn handle the fold boundaries.

5 NumPy: the foundation

5.1 The ndarray

NumPy provides one central object: the n -dimensional array. Unlike a Python list, every element has the same type and the whole thing sits in one contiguous block of memory. That is what makes it fast, and it is why every other library in the stack is built on it.

```
import numpy as np
a = np.array([[1, 2, 3],
              [4, 5, 6]])
print(a.shape, a.dtype)
print(a.sum(axis=0)) # down the columns
print(a.sum(axis=1)) # across the rows
```

```
(2, 3) int64
[5 7 9]
[ 6 15]
```

The `axis` argument confuses everybody at first. The rule: `axis=0` collapses the rows (giving one value per column); `axis=1` collapses the columns (one value per row). Say “the axis I name is the one that disappears”.

5.2 Vectorisation, and why it matters

Consider computing the mean squared error between predictions and truth. Written as a loop:

```
total = 0.0
for i in range(len(y)):
    d = y[i] - yhat[i]
    total += d * d
mse = total / len(y)
```

Written with NumPy:

```
mse = np.mean((y - yhat) ** 2)
```

The second is shorter, but the important difference is speed. Timed on one million elements:

```
loop : 162.9 ms
numpy : 1.85 ms
ratio : 88x
```

The loop runs in the Python interpreter, one element at a time, with all the overhead that implies. The NumPy version dispatches to compiled C which walks a contiguous block. The gap widens with array size.

Key idea

You have just written the **Mean Squared Error**, the first loss function of Unit II. Every optimiser in Unit III exists to make that number smaller. The whole course is, in a sense, an elaboration of `np.mean((y - yhat)**2)`.

MSE

Three predictions. True values $y = (3, 1, 4)$, predictions $\hat{y} = (2.5, 1.5, 4.5)$.
Errors: $y - \hat{y} = (0.5, -0.5, -0.5)$.

$$\text{MAE} = \frac{|0.5| + |-0.5| + |-0.5|}{3} = \frac{1.5}{3} = 0.5$$

$$\text{MSE} = \frac{0.5^2 + (-0.5)^2 + (-0.5)^2}{3} = \frac{0.75}{3} = 0.25$$

$$\text{RMSE} = \sqrt{0.25} = 0.5$$

Verify:

```
y = np.array([3.0, 1.0, 4.0])
yhat = np.array([2.5, 1.5, 4.5])
print("MAE ", np.mean(np.abs(y - yhat)))
print("MSE ", np.mean((y - yhat)**2))
print("RMSE", np.sqrt(np.mean((y - yhat)**2)))
```

```
MAE 0.5
MSE 0.25
RMSE 0.5
```

MAE and RMSE agree here only because every error has the same magnitude. Change the first prediction to $\hat{y}_1 = 0$, so the errors become $(3, -0.5, -0.5)$, and recompute:

$$\text{MAE} = \frac{3 + 0.5 + 0.5}{3} = 1.333, \quad \text{RMSE} = \sqrt{\frac{9 + 0.25 + 0.25}{3}} = \sqrt{3.167} = 1.780$$

RMSE is now a third larger than MAE, because squaring punishes the one big error far more than the two small ones. That asymmetry is exactly why the choice of loss function matters — and it is the subject of Unit II.

5.3 Broadcasting and boolean masks

Two more NumPy ideas you will use constantly.

Broadcasting lets arrays of different shapes combine, by stretching the smaller one:

```
X = np.array([[1., 2.], [3., 4.], [5., 6.]])
col_means = X.mean(axis=0) # shape (2,)
print(col_means)
print(X - col_means) # (3,2) - (2,) works: broadcast down the rows
```

```
[3. 4.]
[[-2. -2.]
 [ 0.  0.]
 [ 2.  2.]]
```

You have just centred a dataset — the first half of what **StandardScaler** does.

Boolean masks select without loops:

```
v = np.array([3, 8, 1, 9, 4])
print(v > 4)
print(v[v > 4])
print((v > 4).sum()) # True counts as 1: how many exceed 4
```

```
[False True False True False]
[8 9]
2
```

6 pandas: tabular data

Where NumPy gives you a grid of numbers, pandas gives you a table with column names, mixed types, and a notion of missing data. That is what real datasets look like.

6.1 The first four calls, every time

Before you model anything, run these four. They take five seconds and answer most of the questions you would otherwise discover painfully later.

```
import pandas as pd
from sklearn.datasets import load_iris

d = load_iris(as_frame=True)
df = d.frame
df.head(3)
```

```
   sepal length (cm) sepal width (cm) petal length (cm) petal width (cm) target
0  5.1  3.5  1.4  0.2  0
1  4.9  3.0  1.4  0.2  0
2  4.7  3.2  1.3  0.2  0
```

```
df.describe().round(2)
```

```
   sepal length (cm) sepal width (cm) petal length (cm) petal width (cm) target
count  150.00  150.00  150.00  150.00  150.00
mean    5.84    3.06    3.76    1.20    1.00
std     0.83    0.44    1.77    0.76    0.82
min     4.30    2.00    1.00    0.10    0.00
25%     5.10    2.80    1.60    0.30    0.00
50%     5.80    3.00    4.35    1.30    1.00
75%     6.40    3.30    5.10    1.80    2.00
max     7.90    4.40    6.90    2.50    2.00
```

Read that table rather than skipping it. Petal length has a standard deviation of 1.77 while petal width has 0.76 — more than double. Sepal width ranges over 2.4 cm while petal length ranges over 5.9 cm. Those differences will matter enormously in Section 7.

The other two calls:

```
df.info() # dtypes and non-null counts: finds text stored as numbers
df.isna().sum() # missing values per column
```

Key idea

`head`, `info`, `describe`, `isna().sum()`. Build the habit now. Unit IV is the disciplined, systematic version of this reflex, and lab experiment 2 assesses it.

6.2 Selecting, creating and grouping

```
df["petal ratio"] = df["petal length (cm)"] / df["petal width (cm)"]
print(df.groupby("target")["petal ratio"].mean().round(2))
```

```
target
0  6.91
1  3.24
2  2.78
Name: petal ratio, dtype: float64
```

That one line has done something useful: it shows setosa (target 0) has petals more than twice as long as they are wide relative to the others, which is precisely why it separates so cleanly in the figure below. Creating a feature like this from domain reasoning is called **feature engineering**, and it is Unit IV.

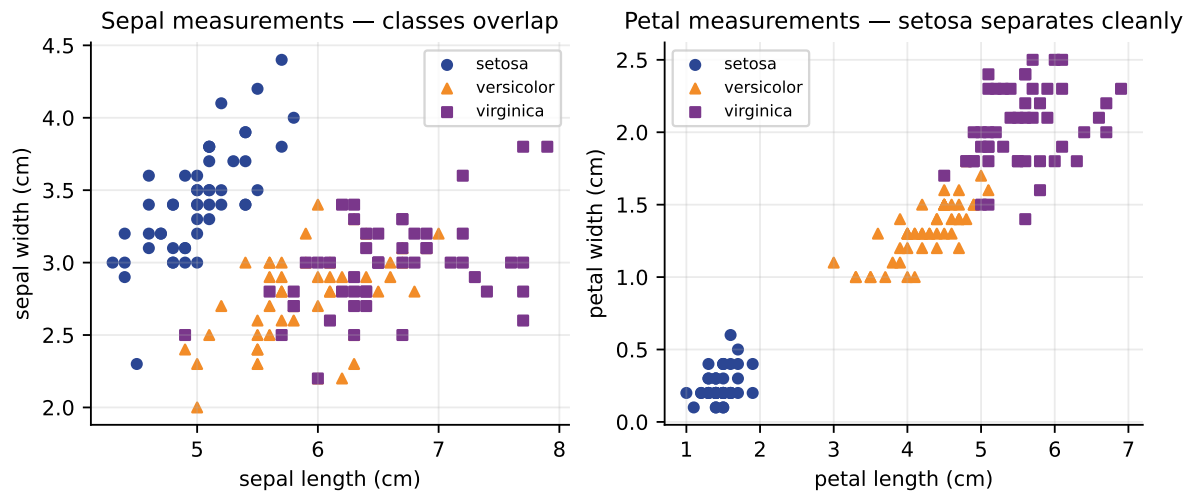


Figure 4: The Iris data in two different feature pairs. Sepal measurements (left) leave the three species overlapping. Petal measurements (right) separate setosa completely and leave only versicolor and virginica touching. Choosing which features to use is often worth more than choosing which model to use.

7 scikit-learn: one API for every model

7.1 The estimator interface

Every model in scikit-learn — and there are hundreds — exposes the same three methods:

```
est.fit(X, y) # learn from data
est.predict(X) # apply to new data
est.score(X, y) # evaluate
```

Every transformer exposes:

```
tr.fit(X) # learn the transformation (e.g. the mean and sd)
tr.transform(X) # apply it
tr.fit_transform(X) # both, for convenience - on TRAINING data only
```

This uniformity is the library's central design decision, and it pays off immediately: swapping a k -nearest-neighbours classifier for a support vector machine changes one line and nothing else. Learn the interface once here and every model in Units V–VII is already familiar.

7.2 An entire workflow in twelve lines

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.pipeline import make_pipeline

X, y = load_iris(return_X_y=True)
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=0, stratify=y)

model = make_pipeline(StandardScaler(), KNeighborsClassifier(n_neighbors=5))
model.fit(X_train, y_train)
print("test accuracy:", model.score(X_test, y_test))
```

```
| test accuracy: 0.9736842105263158
```

Each argument is doing a job:

- `test_size=0.25` holds back 38 of the 150 flowers.
- `random_state=0` fixes the shuffle so you and I get the same split. Without it, every run gives a different answer and nothing is reproducible.
- `stratify=y` keeps the class proportions the same in both halves. Without it a random split can, by bad luck, give you a test set with too few of one species.
- `make_pipeline` chains the scaler and the model into one object, so the scaler is fitted *inside* the training portion only. This is the leak-proof construction from Section 4.5.

Look at what the model actually got right and wrong:

```
from sklearn.metrics import confusion_matrix
print(confusion_matrix(y_test, model.predict(X_test)))
```

```

                pred setosa pred versicolor pred virginica
true setosa    13  0  0
true versicolor  0 13  0
true virginica  0  1 11

```

One error in 38: a virginica called versicolor. That is exactly the pair the figure showed touching, so the model failed precisely where the data is genuinely ambiguous. A confusion matrix tells you *which* mistakes are made; an accuracy figure never does.

7.3 Why the pipeline is not optional

Section 4.5 argued for pipelines on correctness grounds. Here is the other argument: without scaling, distance-based models can fail completely.

We construct a deliberately unfair dataset — one feature that determines the class, on a range of 0–1, and one pure-noise feature on a range of 0–10 000:

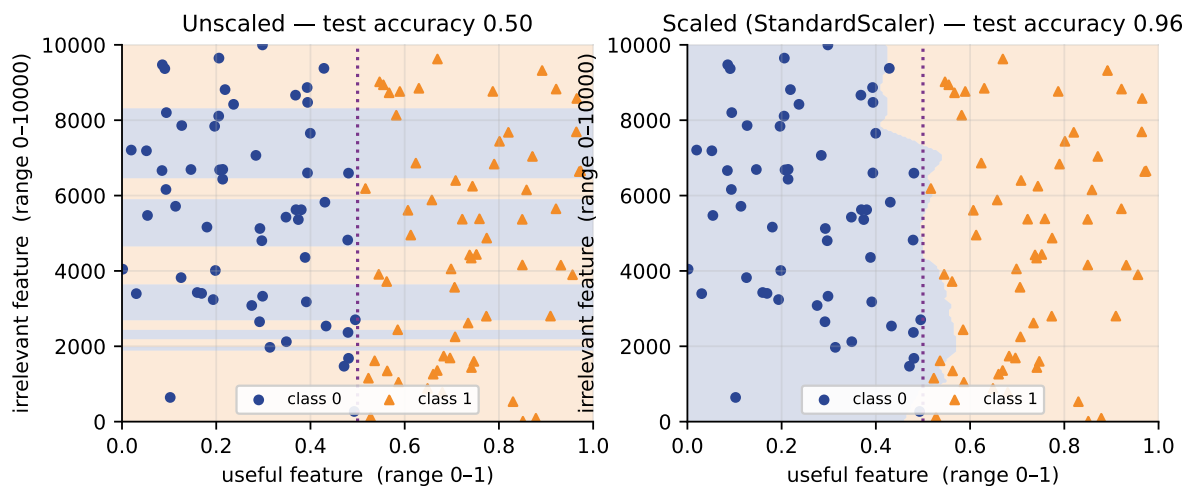


Figure 5: The same KNN classifier on the same data. Left, unscaled: the irrelevant feature’s huge numeric range dominates the distance calculation, the decision regions run horizontally, and accuracy is 0.50 — pure chance. Right, with `StandardScaler` inside the pipeline: the boundary is vertical, following the feature that actually matters, and accuracy is 0.96.

Model	Test accuracy
<code>KNeighborsClassifier(5)</code> on raw features	0.50
<code>make_pipeline(StandardScaler(), KNeighborsClassifier(5))</code>	0.96

The reason is arithmetic. Euclidean distance squares the difference in each feature and adds. A difference of 3000 in the noise feature contributes 9,000,000; a difference of 0.5 in the useful feature contributes 0.25. The useful feature is not slightly outweighed — it is invisible.

Key idea

Any method that measures distance (KNN, SVM with RBF kernel, K-means) or that uses gradient descent (Units II–III) needs features on comparable scales. Tree based methods do not, because they split one feature at a time. Knowing which camp a method is in tells you whether scaling is optional.

7.4 Choosing a hyperparameter

k in KNN is a **hyperparameter**: a setting you choose, not a value the model learns. Choose it by measurement, not by taste.

```
for k in (1, 3, 5, 7, 9, 15, 25, 50):  
    m = make_pipeline(StandardScaler(), KNeighborsClassifier(k))  
    print(f"k={k:<3} cv accuracy {cross_val_score(m, X, y, cv=10).mean():.3f}")
```

```
k=1 cv accuracy 0.953  
k=3 cv accuracy 0.953  
k=5 cv accuracy 0.953  
k=7 cv accuracy 0.953  
k=9 cv accuracy 0.953  
k=15 cv accuracy 0.960  
k=25 cv accuracy 0.947  
k=50 cv accuracy 0.880
```

Two lessons. First, performance is flat from $k = 1$ to $k = 9$ — on this easy dataset the choice barely matters, and a student who reports “I tuned k and got a big improvement” has probably fooled themselves. Second, at $k = 50$ accuracy collapses: with only 50 flowers per species, a 50-neighbour vote inevitably reaches across class boundaries. Both failure modes (k too small fits noise, k too large smooths across classes) are visible in one table.

7.5 More data or a better model?

A common instinct when a model underperforms is to try a fancier model. Often the right answer is more data — and you can find out which, by plotting a learning curve.

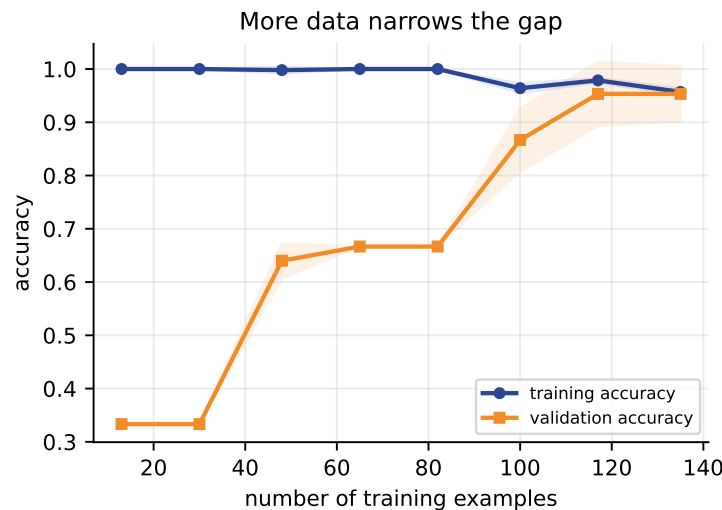


Figure 6: Accuracy against training-set size for KNN on Iris. The two curves have essentially converged by 120 examples, which says that adding more Iris flowers would not help. When the curves are still far apart at the right-hand edge, more data is the cheapest improvement available.

8 Summary

- Machine learning suits problems where you have **examples of the answer** but cannot **state the rule**.
- Mitchell's T , P , E forces you to name the performance measure before you model. Most project failures are P failures.
- Three families: **supervised** (regression, classification), **unsupervised** (clustering, dimensionality reduction, anomaly detection), **reinforcement** (a policy learned from reward).
- The goal is never to fit the training data — it is to generalise. The **gap** between training and test error is your diagnostic.
- A single split on a small dataset is unreliable. Cross-validate, and quote the spread.
- **Data leakage** produced 81% accuracy on pure noise. Put every fitted step inside a **Pipeline**.
- **NumPy** → **pandas** → **scikit-learn**, with one **fit/predict** interface for every model.
- Distance-based and gradient-based methods need scaling. Trees do not.

Next (L2): loss functions. We put a number on how wrong a prediction is — MSE, MAE and Huber for regression, cross-entropy for classification, hinge and triplet for margins. Everything in Units V–VII minimises one of them.

9 Practice questions

Attempt these before reading Section 10.

- Q1.** State Mitchell’s T , P and E for a system that recommends films to a streaming subscriber.
- Q2.** A hospital wants to predict which patients will miss an appointment. Which of the five cautions in Section 2.6 is most likely to bite, and why?
- Q3.** Classify each as supervised regression, supervised classification, unsupervised, or reinforcement:
- (a) predicting tomorrow’s electricity demand;
 - (b) grouping news articles by topic with no labels;
 - (c) deciding whether a transaction is fraudulent, given past confirmed fraud;
 - (d) tuning a thermostat rewarded for comfort and penalised for energy use.
- Q4.** Using the five training flowers from the worked example in Section 3, predict the species of a flower measuring (1.5, 0.3) using 1-NN. Show the distances.
- Q5.** A colleague reports 99% accuracy on a medical test set. What three questions would you ask before believing it?
- Q6.** Rewrite this loop using NumPy, and name the quantity it computes:

```
s = 0
for i in range(len(a)):
    s += abs(a[i] - b[i])
result = s / len(a)
```

- Q7.** Explain, in terms of Euclidean distance, why the unscaled classifier in Section 7 scored exactly 0.50.
- Q8.** In the overfitting table, degree 12 had the lowest training RMSE of all. Why is that not a reason to prefer it?
- Q9.** Your cross-validation gives folds [0.95, 0.93, 0.97, 0.62, 0.96]. What should you investigate, and why is reporting the mean alone inadequate?
- Q10.** Why does `make_pipeline(StandardScaler(), KNN())` give an honest cross-validation score when scaling the whole array first does not?

10 Solutions

- Q1.** T : recommend a film for a given subscriber. E : past viewing and ratings history. P : a stated, measurable quantity such as click-through rate on recommendations, or watch-time of recommended titles. Any defensible P is acceptable *provided it is named explicitly and could actually be measured* — “user satisfaction” is not a performance measure until you say how you would compute it.
- Q2.** Most likely caution 5, **distribution shift**. Appointment-attendance behaviour depends on clinic policy, transport, weather and season, all of which change; a model trained on last year degrades. Caution 4 (explainability) is also defensible if the prediction is used

to allocate scarce appointments, since a patient denied a slot may reasonably demand a reason. Caution 3 deserves a mention too: the cost of wrongly predicting attendance is not symmetric.

- Q3.** (a) supervised regression; (b) unsupervised clustering; (c) supervised classification; (d) reinforcement learning.
- Q4.** Distances from (1.5, 0.3): (1.4, 0.2) $\rightarrow \sqrt{0.01 + 0.01} = 0.14$; (1.3, 0.2) $\rightarrow \sqrt{0.04 + 0.01} = 0.22$; (4.7, 1.4) $\rightarrow \sqrt{10.24 + 1.21} = 3.38$; (4.5, 1.5) $\rightarrow \sqrt{9.0 + 1.44} = 3.23$; (6.0, 2.5) $\rightarrow \sqrt{20.25 + 4.84} = 5.01$. Nearest is (1.4, 0.2) at 0.14, so 1-NN predicts **setosa**.
- Q5.** Reasonable questions: (i) What is the base rate — if 99% of patients are healthy, 99% accuracy is what you get by predicting “healthy” every time. (ii) How was the test set constructed, and did any preprocessing see it? (iii) What is the recall on the positive class — how many sick patients were missed? Also good: how many test cases were there, and is the test set from the same hospital and equipment as deployment?
- Q6.** `result = np.mean(np.abs(a - b))`. It is the **Mean Absolute Error**, which we meet again in Unit II.
- Q7.** Euclidean distance squares each feature’s difference and sums. The noise feature spans 0–10 000, so typical squared differences are of order 10^6 – 10^7 ; the useful feature spans 0–1, contributing at most 1. The sum is therefore determined almost entirely by the noise feature, so the “nearest” neighbours are those with similar noise values, which carry no class information. The result is a coin flip, hence 0.50.
- Q8.** Because training error measures how well the model reproduces data it has already seen, which a sufficiently flexible model can always do perfectly. It says nothing about new data. The degree-12 model’s test RMSE was 40.52 against 1.10 for degree 3 — it is the worst model available, and training error ranked it best.
- Q9.** Investigate the fourth fold. A single fold far below the others suggests either that the folds are not comparable (an unstratified split leaving one fold with an unusual class mix), or that a cluster of genuinely hard or mislabelled cases landed together. Reporting only the mean (0.886) hides the instability; the standard deviation (0.135) is the signal that something is wrong.
- Q10.** Because `cross_val_score` refits the *whole pipeline* inside each fold. The scaler’s mean and standard deviation are computed from that fold’s training rows only, so the held-out rows contribute nothing to the transformation. Scaling the full array first computes those statistics from every row, including the ones about to be used for evaluation — so the model is tested on data whose statistics it has already absorbed.

11 References and further reading

All links were checked on 4 August 2026. Free and no-login resources are marked [free].

Textbooks

- Müller, A. C. and Guido, S. *Introduction to Machine Learning with Python*. O'Reilly, 2016. **Chapter 1** — prescribed text; the Iris walkthrough there is the same one used in these notes.
- Bishop, C. M. *Pattern Recognition and Machine Learning*. Springer, 2006. **Chapter 1, §1.1** — the polynomial curve-fitting example, which is the classical treatment of over-fitting. [free] from Microsoft Research.
- James, G., Witten, D., Hastie, T., Tibshirani, R. and Taylor, J. *An Introduction to Statistical Learning with Python*. Springer, 2023. **Chapters 1–2**. [free] <https://www.statlearning.com/>
- Murphy, K. P. *Probabilistic Machine Learning: An Introduction*. MIT Press, 2022. **Chapter 1** — an excellent taxonomy of learning problems. [free] <https://probml.github.io/pml-book/book1.html>
- VanderPlas, J. *Python Data Science Handbook*. O'Reilly. [free] in full online — <https://jakevdp.github.io/PythonDataScienceHandbook/>

Online courses

- **NPTEL — Introduction to Machine Learning**, Prof. Balaraman Ravindran, IIT Madras. The closest match to this syllabus, in Indian academic idiom, and free to audit. [free] <https://nptel.ac.in/courses/106106139>
Course preview and enrolment: https://onlinecourses.nptel.ac.in/noc23_cs18/preview
Prof. Ravindran's index of NPTEL courses: <https://www.cse.iitm.ac.in/~ravi/nptel-courses/>
- **SWAYAM** — the national portal hosting NPTEL and other Indian university courses. [free] <https://swayam.gov.in/>
- **Google Machine Learning Crash Course** — short, practical, with interactive exercises. Good for the vocabulary in this lecture. [free] <https://developers.google.com/machine-learning/crash-course>
- **Kaggle Learn — Intro to Machine Learning** — four hours, entirely hands-on, runs in the browser with no installation. [free] <https://www.kaggle.com/learn/intro-to-machine-learning>

Videos

- StatQuest with Josh Starmer, *A Gentle Introduction to Machine Learning* (about 12 minutes). The clearest short explanation of training versus testing that exists. [free] https://youtu.be/Gv9_4yMHFI
- StatQuest, *Machine Learning Fundamentals: Bias and Variance*. Watch this before Unit V. [free] <https://youtu.be/EuBBz3bI-aa>

- StatQuest, *Machine Learning Fundamentals: Cross Validation* — covers Section 4 of these notes in six minutes. [free] <https://youtu.be/fSytzGwwBVw>
- StatQuest full video index, organised by topic, useful all semester. [free] <https://statquest.org/video-index/>

Documentation and interactive explainers

- scikit-learn, *Getting Started*. [free] https://scikit-learn.org/stable/getting_started.html
- scikit-learn, *Common Pitfalls and Recommended Practices* — the official treatment of leakage and of `random_state`. Read this one properly; it covers Section 4.5 in more depth. [free] https://scikit-learn.org/stable/common_pitfalls.html
- scikit-learn, *Cross-validation*. [free] https://scikit-learn.org/stable/modules/cross_validation.html
- NumPy, *NumPy: the absolute basics for beginners*. [free] https://numpy.org/doc/stable/user/absolute_beginners.html
- pandas, *10 minutes to pandas*. [free] https://pandas.pydata.org/docs/user_guide/10min.html
- **MLU-Explain** — visual, interactive explanations of train/test splits, bias-variance and cross-validation from Amazon’s machine learning university. Genuinely worth an hour. [free] <https://mlu-explain.github.io/>

Articles

- Brownlee, J. *Data Leakage in Machine Learning*, Machine Learning Mastery. A practical companion to Section 4.5. [free] <https://machinelearningmastery.com/data-leakage-machine-learning/>

Sources: Müller and Guido, *Introduction to Machine Learning with Python*, Ch. 1.; Bishop, *Pattern Recognition and Machine Learning*, Ch. 1.; Murphy, *Probabilistic Machine Learning: An Introduction*, MIT Press, 2022, Ch. 1.; Mitchell, *Machine Learning*, McGraw Hill, 1997, Ch. 1.; Scikit-learn documentation, accessed 2026-08-04.